赵新佳

助理研究员 · 中国科学院大连化学物理研究所

zhaoxinjia@dicp.ac.cn · ORCID 0000-0001-5101-7273 · zhaoxinjia.github.io

教育与工作经历

2023.02 – 至今	助理研究员，中国科学院大连化学物理研究所（卿光焱 研究员团队）
2021.01 – 2023.02	博士后，中国科学院物理研究所（合作导师：陆兴华 研究员）
2015.09 – 2021.01	博士，凝聚态物理，中国科学院物理研究所 表面物理国家重点实验室（导师：陆兴华 研究员）
2011.09 – 2015.07	学士，物理学，吉林大学 物理学院

研究方向

从事纳米孔单分子检测器件与生物医学分析方法研究，围绕蛋白质翻译后修饰、糖链结构与药物–核酸相互作用，开展分子识别方法、检测平台与机理研究。

- **纳米孔分子识别。**提出纳米孔工程化 PIToN 模型，实现蛋白质瓜氨酸化修饰的亚道尔顿级单分子辨识；发展面向中性糖链、唾液酸连接异构体及顺铂–DNA 加合物检测的新方法。
- **单分子操控平台。**研制固态纳米孔核心检测模块，发展分子乒乓与重复捕获技术，实现同一分子的多次测量，降低单次穿孔事件的随机波动，提高分子表征的一致性。
- **机制与理论。**结合分子动力学模拟与有限元模拟，解析孔道界面相互作用、分子运输与离子电流信号之间的关系，指导孔道设计与检测条件优化。

代表性论文

（共发表论文 24 篇、接收 1 篇；# 共同第一作者，⋆ 通讯作者；分区为中科院大类分区）

[1] **Xinjia Zhao**#, Jiarong Zhang#, Juanjuan Zhao#, Wei Yong, Zeyuan Zhao, Minmin Li, Yuchen Cao, Wenjing Sun, Jing Wang, Dongdong Wang, Jiaqi Li, Kuoxi Xu, Yangyang Bian, Guangyan Qing⋆. Sub-dalton discrimination of citrullination by a semi-rationally designed aerolysin nanopore. *Nat. Commun.*, 2026, 接收（综合 1 区 TOP）

[2] **Xinjia Zhao**, Yahui Zhang, Guangyan Qing⋆. Nanopore toward genuine single-molecule sensing: Molecular ping-pong technology. *Nano Lett.*, 2025, 25, 3692–3706.（材料 2 区 TOP）

[3] **Xinjia Zhao**, Haijuan Qin, Mingliang Tang, Xiaoyu Zhang, Guangyan Qing⋆. Nanopore: Emerging for detecting protein post-translational modifications. *Trends Anal. Chem.*, 2024, 173, 117658.（化学 1 区 TOP）

[4] Wenqi Lu#, **Xinjia Zhao**#, Minmin Li, Yuting Li, Chen Zhang, Yuting Xiong, Jiaqi Li, Han Zhou, Xianlong Ye, Xiaonong Li, Jing Wang, Xinmiao Liang⋆, Guangyan Qing⋆. Precise structural analysis of neutral glycans using aerolysin mutant T240R nanopore. *ACS Nano*, 2024, 18, 12412–12426.（材料 1 区 TOP）

[5] Dongdong Wang, **Xinjia Zhao**⋆, Minmin Li, Jiarong Zhang, Yingying Lian, Guangyan Qing⋆. Multiscale theoretical calculations drive mechanism analysis and application expansion of nanopore detection. *Anal. Sens.*, 2026, 6, e202600002.（化学 3 区）

[6] **Xinjia Zhao**, Xiaoyu Chen, Zhuang Mi, Yuru Liu, Wei Li, Xinyan Shan⋆, Xinghua Lu⋆. Revealing differential interaction forces during nanopore DNA sequencing. *J. Phys. Chem. B*, 2021, 125, 5045–5051.（化学 2 区）

[7] **Xinjia Zhao**#, Yuru Liu#, Xiaoyu Chen, Zhuang Mi, Wei Li, Pengye Wang⋆, Xinyan Shan⋆, Xinghua Lu⋆. Detection and characterization of single cisplatin adducts on DNA by nanopore sequencing. *ACS Omega*, 2021, 6, 17027–17034.（化学 2 区）

[8] **Xinjia Zhao**, Ruiping Ma, Ying Hu, Xiaoyu Chen, Ruifen Dou, Ke Liu, Chengjun Cui, Huajie Liu, Qian Li, Dun Pan, Xinyan Shan⋆, Lihua Wang⋆, Chunhai Fan, Xinghua Lu⋆. Translocation of tetrahedral DNA nanostructures through a solid-state nanopore. *Nanoscale*, 2019, 11, 6263–6269.（化学 2 区）

发明专利（已授权）

[1] 一种制备绝缘基座的模具、方法及基座在分子探测中的应用，专利号 ZL201811088465.1。发明人：赵新佳，单欣岩，陆兴华。

[2] 基于纳米孔测序的力谱分析方法和分析装置，专利号 ZL202110598008.2。发明人：赵新佳，陈晓钰，米壮，单欣岩，陆兴华。

[3] 控制生物分子在纳米孔中运动的装置和方法，专利号 ZL201911053046.9。发明人：陈晓钰，赵新佳，米壮，贺雅茹，单欣岩，陆兴华。

发明专利申请

[4] 一种纳米孔微流腔及其应用，申请号 CN202010947474.2。发明人：陈晓钰，赵新佳，米壮，单欣岩，陆兴华。

[5] 高通量筛选生物分子的装置及其应用，申请号 CN202010946762.6。发明人：陈晓钰，赵新佳，米壮，单欣岩，陆兴华。

主持与参与科研项目

2024.01 – 2025.12 主持，大连化物所–大连医科大学附属第一医院医工联合创新基金（DMU-1&DICP UN202307）：聚合物修饰的多孔钛合金支架及其在大段骨缺损修复中的应用，50 万元

2018.01 – 2020.12 参与，国家自然科学基金重大研究计划（91753136）：利用纳米孔技术探测生物大分子动态修饰研究，70 万元

2016.01 – 2019.12 参与，国家自然科学基金面上项目（31571012）：基于 DNA 自组装标记的纳米孔测序新技术研究，66 万元

获奖与荣誉

2025 中国分析测试协会科学技术奖（CAIA）一等奖

2024 辽宁省“兴辽英才”博士后储备计划

2023 中国科学院大连化学物理研究所高层次人才（青年才俊）引进计划

学术服务与会员

2026 – 至今 Sigma Xi（科学研究荣誉学会）正式会员（Full Member，经提名选举）

2024 – 至今 辽宁省细胞生物学学会心血管疾病医工融合专业委员会 委员

其他论文

（代表性论文之外的 17 篇，按发表年份倒序；编号接续上文）

[9] Xiaoyu Chen, Jiao-ao Sun, **Xinjia Zhao**, Zhuang Mi, Haitao Tang, Kaikai Chen, Xinyan Shan, Xinghua Lu*. Quartz-supported suspended membrane for ultra-sensitive solid-state nanopore with sub-1 pF capacitance. *Adv. Funct. Mater.*, **2026**, 36, e77522.

[10] Jiaqi Li, **Xinjia Zhao**, Minmin Li, Wei Yu, Yuting Li, Chen Zhang, Hongzhi Cao, Kan Zhong, Jing Wang, Yuchen Cao, Xiaonong Li, Han Zhou, Guoxiong Wang, Xinmiao Liang*, Guangyan Qing*. Single-feature identification of α2–8 linked sialoglycans using engineered aerolysin nanopores: A paradigm for glycan linkage analysis. *J. Am. Chem. Soc.*, **2026**, 148, 10649–10662.

[11] Jing Wang, Minmin Li*, Chen Zhang, **Xinjia Zhao**, Yuting Xiong, Yuchen Cao, Dongdong Wang, Xiaonong Li, Xinmiao Liang*, Guangyan Qing*. Identification of isomerically diverse ginsenosides using engineered aerolysin nanopore via non-translocation blockade sensing. *Angew. Chem. Int. Ed.*, **2025**, 64, e202506741.

[12] Wenjing Sun, Xinmiao Zhao, **Xinjia Zhao**, Lingkai Meng, Mingliang Tang, Jiaqi Li, Yongxin Chang, Yuting Xiong, Hao Wang, Jinghua Chen*, Guangyan Qing*. Significantly enhanced capture efficiency of cell-imprinted material for circulating tumor cells via a flexible and ultra-strong double-armed phenylboronic acid design. *Biomaterials*, **2025**, 322, 123397.

[13] Yahui Zhang, **Xinjia Zhao**, Guangyan Qing*. Smart polymers advance analytical chemistry. *Trends Anal. Chem.*, **2025**, 187, 118198.

[14] Yongxin Chang, Juan Shao, **Xinjia Zhao**, Haijuan Qin, Yanqing Du, Junrong Li, Qiongya Li, Wenjing Sun, Guoxiong Wang, Guangyan Qing*. Precise AIE-based ternary co-assembly for saccharide recognition and classification. *Adv. Sci.*, **2024**, 11, 2405613.

[15] Minmin Li, Jing Wang, Chen Zhang, **Xinjia Zhao**, Yuting Xiong, Yuchen Cao, Dongdong Wang, Xiaonong Li, Xinmiao Liang*, Guangyan Qing*. Single-molecule identification and quantification of steviol glycosides with a deep learning-powered nanopore sensor. *ACS Nano*, **2024**, 18, 25155–25169.

[16] Zhuang Mi, **Xinjia Zhao**, Xiaoyu Chen, Yichao Wang, Xinyan Shan*, Kaikai Chen*, Xinghua Lu*. Ten thousand recaptures of a single DNA molecule in a nanopore and variance reduction of translocation characteristics. *ACS Nano*, **2024**, 18, 23243–23252.

[17] Zhuang Mi, Xiaoyu Chen, **Xinjia Zhao**, Haitao Tang, Wenyu Wang, Xinyan Shan, Xinghua Lu*. High-precision high-speed nanopore ping-pong control system based on field programmable gate array. *Rev. Sci. Instrum.*, **2024**, 95, 073202.

[18] Wenjing Sun, Xin You, **Xinjia Zhao**, Xiaoyu Zhang, Chunhui Yang, Fusheng Zhang, Jiaqi Yu, Kaiguang Yang, Jixia Wang, Fangfang Xu, Yongxin Chang, Boxin Qu, Xinmiao Zhao, Yuxuan He, Qi Wang*, Jinghua Chen*, Guangyan Qing*. Precise capture and dynamic release of circulating liver cancer cells with dual-histidine-based cell imprinted hydrogels. *Adv. Mater.*, **2024**, 36, e2402379.

[19] Jie Xiao, Muyu Cong, Minmin Li, Xin Zhang, Yahui Zhang, **Xinjia Zhao**, Wenqi Lu, Zhimou Guo, Xinmiao Liang*, Guangyan Qing*. Self-assembled nanoporous metal–organic framework monolayer film for osmotic energy harvesting. *Adv. Funct. Mater.*, **2024**, 34, 2307996.

[20] Yuting Xiong#, Minmin Li#, Yuchen Cao, Zan Li, Yongxin Chang, **Xinjia Zhao**, Guangyan Qing*. Nanofluidic device for detection of lysine methylpeptides and sensing of lysine methylation. *Anal. Chem.*, **2023**, 95, 7761–7769.

[21] Yue Qin#, Xiaoyu Zhang#, Yanling Song, Bowen Zhong, Lu Liu, Dongdong Wang, Yahui Zhang, Wenqi Lu, **Xinjia Zhao**, Zhiqi Jia, Minmin Li, Lihua Zhang*, Guangyan Qing*. A highly sensitive nanochannel device for the detection of SUMO1 peptides. *Chem. Sci.*, **2023**, 14, 8360–8368.

[22] Hao Dong, Yueyi Wang, Xiangqian Tang, **Xinjia Zhao**, Yang An, Wenyu Wang, Jianmei Li, Xinyan Shan, Xinghua Lu*. Geometrical and magnetic properties of small titanium and chromium clusters on monolayer hexagonal boron nitride. *Phys. Chem. Chem. Phys.*, **2023**, 25, 6079–6088.

[23] Yikun Yao, **Xinjia Zhao**, Xiangqian Tang, Jianmei Li, Xinyan Shan*, Xinghua Lu*. Laser etching of 2D materials with single-layer precision up to ten layers. *J. Laser Appl.*, **2022**, 34, 042051.

[24] Yang An, Xiangqian Tang, Wenyu Wang, Hao Dong, **Xinjia Zhao**, Muyu Wang, Ye Xia, Xinyan Shan*, Xinghua Lu*. Spatially resolved stimulation for the controlled debromination in single molecules on a surface. *ACS Nano*, **2022**, 16, 18592–18600.

[25] Xiaoyu Chen, **Xinjia Zhao**, Ruiping Ma, Ying Hu, Chengjun Cui, Zhuang Mi, Ruifen Dou, Dun Pan, Xinyan Shan*, Lihua Wang, Chunhai Fan, Xinghua Lu*. Ionic current fluctuation and orientation of tetrahedral DNA nanostructures in a solid-state nanopore. *Small*, **2022**, 18, 2107237.